Nithin Salevemula

Pronouns: he/him

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Research Interests

Distributed Algorithms, Game Theory, Rational Distributed Computing, Cryptography, and Quantum Computation.

Education

2024–present **Ph.D., Computer Science & Engineering**, *Indian Institute of Technology Madras*.
Coursework includes Distributed Trust, Distributed Algorithms, Quantum Algorithms and Complexity, Classical Cryptography, Game Dynamics and Incentive-Centered Design. CGPA: 9.0/10

2017–2021 **B.Tech., Computer Science & Engineering**, *Indian Institute of Technology Madras*.
Relevant coursework includes Advanced Graph Algorithms, Randomized Algorithms, Computational Geometry, and Game Theory. CGPA: 7.04/10

Papers

2027 **N. Salevemula** and S. Pai, *"Improved Message Complexity for Fair Leader Election,"* **ICDCN 2027.**

2026 **N. Salevemula** and S. Pai, *"Distributed MIS Algorithms for Rational Agents using Games,"* **Accepted to the International Colloquium on Structural Information and Communication Complexity (SIROCCO 2026).** Preprint available at arXiv:2511.16533.

Current Research Projects

Indian Institute of Technology, Madras

July,2024 – present ***Distributed Algorithms in the network of Rational Nodes***.
We study classical distributed computing problems in a *rational* setting, where nodes are modeled as self-interested agents that may strategically deviate from prescribed algorithms to maximize their individual utility. We focus on *natural utility models* that arise inherently from the problem structure. The objective is to design distributed algorithms that are *rationality resilient* without relying on any monetary transfers or payments external to the network

Advisor : **Dr. Shreyas Pai**, *Associate Professor, Department of Computer Science & Engineering*, IIT Madras (*Personal Web-page*)

Course Projects

Indian Institute of Technology, Madras

Jan,2025 – July,2025 ***Carbon Ledger***, *CS6858 : Distributed Trust*.
Designed and implemented a full-stack carbon credit platform using Solidity and Hedera. Implemented a token-based carbon credit mechanism with role-based permissions(Seller, Buyer, Verifier, Auditor, Admin), enabling secure user onboarding, credit minting, validation, trading, and retirement on-chain. Implemented HBAR-based payment mechanisms for credit purchase and retirement, and built a complete frontend integrating MetaMask authentication, event-driven contract interaction, and decentralized document storage using IPFS for project documentation.

Advisor : **Dr. John Augustine**, *Professor, Department of Computer Science & Engineering*, IIT Madras (*Personal Web-page*)

Research Visits

June–July 2026 ***Rational Distributed Computing***, *Durham University*.
Worked on socially optimal distributed leader election and team formation, as well as self-stabilizing maximal independent set (MIS) algorithms for networks of rational agents.

Host: **Dr. Amitabh Trehan**, Durham University (*Personal webpage*)

Fellowships & Awards

Spring and Fall, 2025 **Star TA Award** for exceptional contribution as a Teaching Assistant in the course *Advanced Graph Algorithms* and *Distributed Algorithms*

Work experience

2021–2023 **Software Development Engineer**, *Housing.com*, Gurgoan.
During my time at Housing.com, I worked extensively with Jetpack Compose and the Android framework while contributing to the development of the Proptiger Android application, which is available on the Play Store. I implemented numerous application features following MVVM architecture, including building user interfaces from design specifications, integrating APIs, wiring intents, and interacting with various Android system components. I also integrated and managed third-party services, including Facebook, Firebase, and UXCam, for analytics and testing purposes. In addition, I resolved several high-priority UI and functional bugs and documented the architecture and implementation of the features I developed for internal reference.

Internships

May 2020 - July 2020 **Summer Technology Intern** , *Samsung Research and Development*, Bengaluru.
During my internship, I worked on the problem of determining whether a device should connect to the 2.4 GHz or 5 GHz frequency band of a router based on the application it is running. I learned and applied reinforcement learning techniques to model and solve this decision-making problem. In addition, I used the ns-3 simulator to build and evaluate the corresponding network model.

May 2019 - July 2019 **Summer Technology Intern** , *Trillbit*, Bengaluru.
I worked on an indoor navigation project that utilized WiFi signal strengths from multiple routers in large indoor environments to estimate user location and assist with navigation. As part of the project, I reviewed relevant research literature and identified feasible approaches for accurate indoor positioning. I developed two Android applications: the first estimated user distance using the inverse-square law of electromagnetic propagation, and the second performed location estimation by training on reference points and applying the K-nearest-neighbour method.

Computer skills

Programming Languages C++, Python, C

DApp Development Smart Contracts in Solidity

Mobile development Android SDK, Kotlin, Jetpack compose

Teaching Assistantship

Fall, 2026 : **CS6855: Multi-Agent Systems**, IIT Madras.

Spring, 2026 : **CS6170: Randomized Algorithms**, IIT Madras.

Fall, 2025 : **CS6851: Distributed Algorithms**, IIT Madras.

Spring, 2025 : **CS6130: Advanced Graph Algorithms**, IIT Madras.

Fall, 2024 : **CS2700: Programming and Data Structures** , IIT Madras.

Extra - Curricular Activities

I actively participate in fitness and sports, including gym training, volleyball, table tennis, frisbee, and skating.

Referees

Dr. Shreyas Pai
Assistant Professor, Department of Computer Science & Engineering
IIT Madras
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